

3 adaptive functions of spleen

Splenomegaly

WHO criteria

Tropical

Hypersplenism & tropical

DD for Splenomegaly

Ⓛ Kidney (MA)

Left lobe of liver

CA stomach

Tail CA pancreas

Splenic plexus - mass

Classical sign of cirrhosis

bounding pulse - (D) vol.

Hipocratic face - sunken eyes

If Hepatitis C infection

↓
treat it

↓
cirrhosis - reversible

TbC → Liver transplantation

Ascites management → Spironolactone
Furosemide

Syndromes

Banding

Felty

Adult still

Not useful for systemic disease

Effect of splenectomy in CML - only pain relief
Very big spleen → x dragging heart

Radionuclide scan

↓
Accurate measurement of spleen

↓
in splenomegaly

Hypo splenism
Asplenism

↓
Surgical resection of spleen, Splenectomy

(not absence of spleen)

Abscopal effect

↓
Systemic disease disappears on removal of spleen

Name

Age / sex

Occupation

Address

Complaints & duration

Symptoms for which the patient has come to hospital

&

Symptoms told by patient after questioning
In chronological order.

H/O presenting complaints

~~H/O~~ ~~Ab~~ The patient was apparently normal _____ days
back, after which he developed

H/O Abdominal pain.

Site -

Onset -

Timing { duration -
 { progression -

Character -

Radiation -

Associated with -

Exacerbating / relieving factors -

Co-morbidities

Site

Visceral abdominal pain

Conducted via sympathetic splanchnic nerves

Deep & poorly localised in the midline

Causes

- Hollow organ distension
- Mesenteric traction
- Excessive smooth muscle contraction

Somatic abdominal pain

Conducted via intercostal nerves

Lateralsised & localised to inflamed area.

Causes

- from Parietal peritoneum
- Abdominal wall

Pain from

Foregut

Stomach, pancreas

Liver, biliary system

Localised above

Upper abdomen

Midgut

Small intestine
Appendix

= Central abdomen

Hindgut

Colon

- Lower abdomen

Pain from

Localised

UnPaired structures - pancreas

- Midline, radiates to the back

paired structures - kidneys

- felt on & radiates to affected side.

Males

Torsion of testis

Females

Pelvic inflammatory disease (PID)
ruptured ovarian cyst
endometriosis
ectopic pregnancy

Onset

Sudden - rapidly progressing
(severe) to become generalised
& (constant)

① Hollow viscus
perforations
↓
Peptic ulcer.

Sudden → (Severe) - Acute pancreatitis
Torsion of cecum/sigmoid

② Ruptured abd.
aortic aneurysm

Rapidly rising
(Severe) - Biliary colic
Renal colic

③ Mesenteric
infarction

Gradual
(mild to mod.) - Peptic Ulcer.

Character

Colicky → Shoot time
eases off
returns
Biliary/renal
colic
(Misnomer)

① Small/large intestinal
- obstruction
② Uterus during labor.

Constant - rapidly rises to peak
Several hrs
gradually resolves

① Biliary colic
② Renal colic

Dull, Vague,
constant,
poorly localised

Infection/
inflammation

① Appendicitis
② Diverticulitis
③ Salpingitis

Gnawing - Peptic Ulcer.

Severity

Excruciating pain - poorly relieved
by opioids

Ischaemic

Severe pain - relieved by
opioids

Bowel infarction/
ruptured abd. aortic
aneurysm

- Acute pancreatitis
- Peritonitis
- 2° to ruptured viscus.

Radiation

- ① Acute cholecystitis — from (R) hypochondrium
(Diaphragmatic irritation) to shoulder (or) interscapular region
- ② Renal colic — From loin to groin & genitalia
- ③ Acute pancreatitis — Central (midline) upper — to the back
↓
relieved by (sitting forward)
- ④ Acute Appendicitis — Central abdominal pain
shifts to (R) iliac fossa
- ⑤ Ruptured / dissecting abd. aortic aneurysm } — Severe back pain
+ Severe abd. pain.
- ⑥ Peptic ulcer — Into the back
- ⑦ Biliary colic — below (R) scapula

Associated features

Nausea, Vomiting, Anorexia → Non specific
any severe pain

Altered bowel habit → Irritable bowel syndrome
↓
diverticulum
Colorectal cancer.

Fig 77M

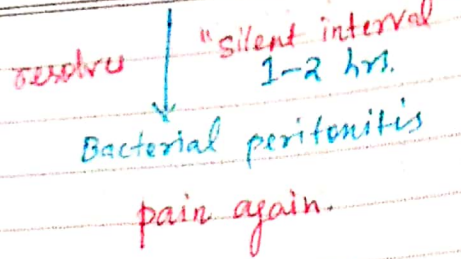
for
Non-alimentary
causes of
abd. pain

Hypotension & tachycardia → Fall. onset of pain

↓
Intra abdominal
or
bleeding from
peptic ulcer / rupture
abd. aortic aneurysm /
ectopic pregnancy

Timing

Chemical peritonitis



Acute appendicitis

DATE

1st periumbilical pain

SOMATIC - VISCERAL

Shifts to right iliac fossa
(localised parietal peritonitis)
SOMATIC

If appendix ruptures

(generalised peritonitis)

Localised pain in right iliac

+ Palpable mass

↓
Appendix abscess.

Pain for hours - days → inflammation

- Acute appendicitis
- Cholecystitis
- diverticulitis
- pancreatitis

Intestinal colic (small intestinal obstruction)

↓
Persistent pain + abd. tenderness

Intestinal ischaemia → EMERGENCY

Peptic ulcer

1/2 - 2 hrs

when? @ night / hungry

remission for wks/months

Exacerbating - stress, spicy food, alcohol, NSAIDs

Relieving - Food, antacids, Vomiting

Biliary colic

4 - 24 hrs

Unpredictable

Attacks can be numbered

eating - can't eat during bout

Renal colic

4 - 24 hrs

After dehydration

Attacks can be no. ed

Alcohol eating - can't eat during bout

Sitting upright.

Acute Pancreatitis

> 24 hrs

After heavy drinking

discrete episode

⊕ - Movements/coughing → inflammation

⊖ - lying still

H/o Painful mouth

- Iron, folate, vit B₁₂, C deficiency
- lichen planus
- aphthous ulcers
- IBD, coeliac diseases
- infective stomatitis.

H/o Heartburn

HOT, burning character

Retrosternal site

Upward radiation

ppted by lying flat / bending forward,

asso. with waterbrash (due to reflex salivation)
taste of acid

H/o dyspepsia

Pain / discomfort in the upper abdomen.

Types

- reflux like → heartburn predominant dyspepsia
- ulcer like → relieved by food / antacids
- dysmotility like → Nausea, bloating, belching, premature satiety.

FAT INTOLERANCE

Common in all
causes of dyspepsia.

POINTING
SIGN

Pt points a
single localised
site in epigastrium

H/O Abdominal distension

Onset

duration

progression

Associated with leg swelling / facial puffiness / abd pain

H/O Vomiting

character

content

projectile / not

Associated with nausea / dyspepsia / abd pain.

before / after food.

Non GI causes

- ↑ sed ICT
- Vestibular disorders
- DKA
- Addison's
- hypercalcemia
- renal / liver failure
- Pregnancy
- drugs → Alcohol
Opioids
Antidepressants
digoxin
cytotoxic agents

Abdominal distension

Fat (obesity)
Flatus (bowel obstruction)
faeces (constipation)
fetus
fluid (Ascites)
full bladder
functional. (Irritable bowel syndrome)
distended during day } fluctuating.
resolves during night }

Nausea → Sensation of feeling sick.

Vomiting → Expulsion of gastric contents via the mouth.

- ① UGI diseases → N, V, + abd. pain
- ② Dyspepsia → Only N, no V, ↑ sed ICT → only V NO N
- ③ Pyloric stenosis → Projectile Vomiting of large volume of gastric contents.
(gastric outlet obstruction)
- ④ Obstruction distal to pylorus → Bile-stained vomit
- ⑤ distal ^{small} bowel / Colonic obstruction → Faeculent Vomiting of small bowel contents.
- ⑥ Peritonitis → Small volume but persistent
- ⑦ Severe pain → ppt. Vomiting
(MT renal colic biliary colic)
- ⑧ Self-induced Vomiting → Anorexia nervosa → wt. loss
Bulimia → wt. maintained or ↑ sed.

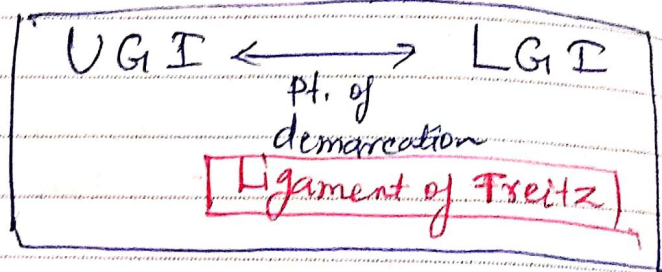
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H/o hematemesis

No. of episodes

quantity

Asso. with melaena



H/o dark coloured stools

black-shiny / black-matt / not.

Streaked / frank blood

Hematemesis - Vomiting of blood (due to UG)

causes - Oesophageal varices → pool ^{blood} up in mouth
↓
actively vomited.

- Mallory-Weiss tear in lower Oesophagus → Forceful retching } → Only pt vomited forcefully several times.

Malena due to UGI bleed

("Passage of Black, tarry, shiny stools with char. foul odour")

causes Peptic ulcer (MCC)

Hematemesis

Vomiting of blood

UGI bleed

Causes

① Oesophageal varices → blood pool up in mouth
in cirrhosis
↓
actively vomited

② Mallory-Weiss tear
in lower oesophagus

Forceful retching

↓
Only when pt vomit forcefully
several times

③ Peptic ulcer - MCC.

④ Erosive gastritis - due to excessive alcohol ingestion.

⑤ Oesophageal / gastric cancer

⑥ gastric angiodysplasia (Dieulafoy lesion)

LGI bleed

Hematochezia (Fresh rectal bleeding)

causes

① Hemorrhoids

② Anal fissures

③ Colorectal cancer

④ Colonic polyps

⑤ Ischaemic colitis

⑥ Colonic angiodysplasia

⑦ IBD

⑧ complicated diverticulum.

Melena

Black, tarry, shiny
stools with char. foul odour.

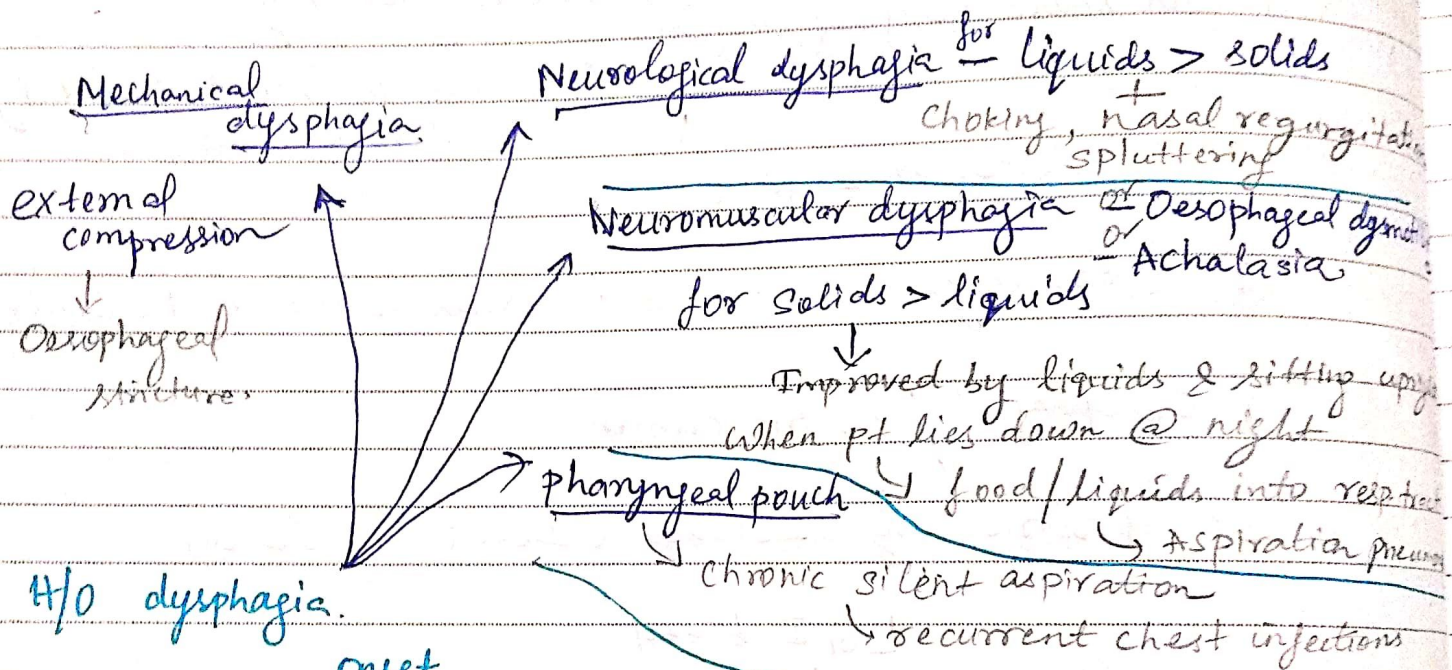
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score
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H/O Jaundice

Duration

colour of urine & stool

local treatment?



onset

progressive/intermittent

To solids/liquids/both

level of sticking of food

any regurgitation of food/fluid

also with - pain (odynophagia) / heart burn / weight loss

Oesophageal cancer

"Pain on swallowing"

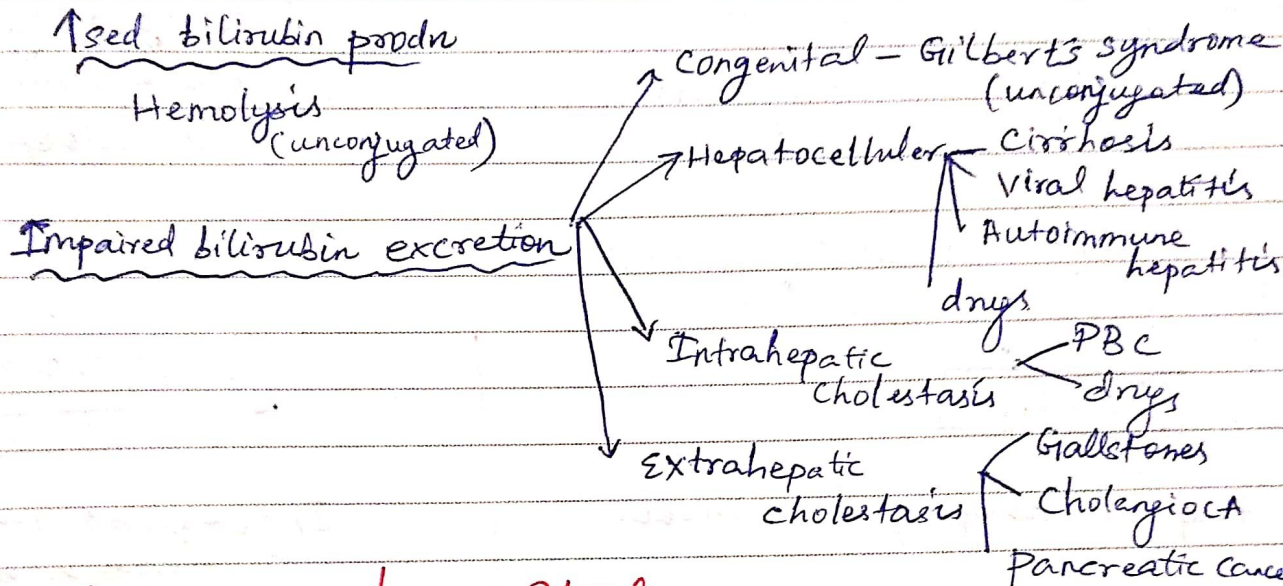
causesOesophageal ulceration/oesophagitis.
due to GERD / oesophageal candidiasis.Indicates - intact mucosal sensation

So X Oesophageal cancer.

JAUNDICE

Yellowish discolouration of skin, sclera & mucous membranes caused by hyperbilirunemia ($2.5 - 3 \text{ mg/dL}$)
 $[2.92 \text{ mg/dL}]$

Causes



JAUNDICE	Urine	Stool
Hemolytic/ Gilbert's	Normal (Acholuric)	Normal
Hepatic	Dark	Normal
Obstructive	Dark	pale

+ pruritis
 + Pain (if due to gall stones)
 Painless (if due to pancreatic CA/ cholangiocA)

H/o loss of weight
H/o loss of appetite.

rapid wt loss over days
↓ loss of body fluid due to vomiting diarrhoea dizziness

H/o altered bowel habit

Diarrhoea

Frequent passage of loose stools

(N) 3/day to 1/3 days

Steatorrhoea

causes disease
chronic pancreatitis
CF causing pancreatic insufficiency

- Fat malabsorption
- greasy, bulky
- pale
- float - can't flush away

High Vol. diarrhoea (> 1L/day)

Stool water content ↑

SECRETORY

Intestinal infection/inflammation

If pt fasts, osmotic stops but secretory doesn't

Acute infective - Norovirus, salmonella, Clostridium difficile

Chronic infective - Giardiasis, Amoebiasis, Cryptosporidiosis

Bloody diarrhoea - Infective Colitis, IBD, Colonic ischaemia

OSMOTIC

malabsorption
laxative abuse
dysmotility

(DM - autonomic neuropathy)

Constipation

Infrequent passage of hard stools

causes

I.B.S

hypothyroidism

hypercalcaemia

colorectal cancer

immobility (stroke)

drugs (Iron/opioids)

Absolute constipation

No bowel mvt

No flatus

↓ Intestinal obstruction

+ pain

+ vomiting

+ abd. distension

Low Vol. diarrhoea

I.B.S.

rectal inflammation
↑ constipation

Tenesmus

Sensation of needing to defecate - but rectum empty

H/O mass in abdomen

size, change in size?
onset, progression
asso. with pain.

DATE

H/O oliguria

H/O hematuria

H/O fever

H/O swelling of legs

H/O bleeding tendencies

Past history

H/O previous similar episodes

H/O jaundice, blood transfusion

H/O TB, DM, HT, CAD, thyroid disease

H/O exposure to STD

H/O previous abdominal surgeries/hospitalization